

Disambiguating “Directed Selection” Across the Trilogy

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This work derives from and formalizes commitments across the CKS theory trilogy: “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026), and “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026). It does not introduce new axioms. Its sole contribution is to establish, with precision, the meaning of the term *directed selection* as it appears across all three papers—showing that one governance-authorized DNA change mechanism operates in three contexts with different source content, not three different mechanisms.

Abstract

The term “directed selection” appears across the CKS theory trilogy in three distinct contexts: implicitly in Paper 1 as an exercise of the modify right over substrate content, explicitly in Paper 2 as the named governance-controlled DNA evolution mechanism operating on intra-Self source content, and in Paper 3 as the same mechanism applied to DNA content absorbed from a contributing Self’s aspects during a Full Aspect Integration (FAI) event. A potential adversary could treat these three usages as separate inventions—claiming that Paper 3’s DNA absorption constitutes a novel evolution mechanism distinct from Paper 2’s directed selection. This note forecloses that reading. Directed selection throughout the trilogy is one mechanism: governance-authorized modification of DNA-layer substrate content, producing versioned DNA change records under human authority. What varies across the three contexts is the source of the content under consideration—unspecified in Paper 1, intra-Self in Paper 2, and inter-Self in Paper 3 via the absorption step of FAI. The mechanism, the authorization requirement, and the record type are identical across all three contexts. This note opens the T2 evolution-concept disambiguation subseries and establishes the pattern those notes will follow: Paper 3’s evolution concepts extend Paper 2’s mechanisms to accept inter-Self source content; they do not introduce new mechanisms.

1. Why this disambiguation is needed

The CKS trilogy reuses several terms at progressively larger coordination scopes. For most terms, the reuse is visible and the scope expansion is explicitly signaled. “Directed selection” presents a more subtle case. Paper 1 does not use the term at all; Paper 2 names and defines it as the primary evolution mechanism for governance-controlled DNA change; Paper 3 introduces DNA absorption through FAI and describes it as operating through Paper 2’s DNA evolution mechanism—but does not pause to make explicit that absorption *is* directed selection with a different source.

That gap creates an adversary opportunity. An adversary could argue that Paper 3’s DNA absorption is a novel mechanism—one that adds inter-organizational content ingestion, provenance-carry-over at the perimeter, and governance-configured selective filtering—and that this collection of properties makes it architecturally distinct from Paper 2’s directed selection, which sources improvement proposals from the proposing Self’s own action-layer records. If that argument succeeded, it could be used to claim novelty in Paper 3’s absorption protocol and to sever the inheritance chain from Paper 2 that grounds DNA absorption in already-established governance authority.

The disambiguation this note provides closes that gap. The closing argument is simple: the mechanism is identical; the source differs. Once that is precise, the inheritance chain is unambiguous, no new governance authority is required for absorption, and the trilogy’s evolution architecture forms a single continuous arc from Paper 1’s modify right through Paper 2’s directed selection to Paper 3’s absorption-as-directed-selection.

2. The term across three paper contexts

2.1 Paper 1: Implicit directed selection via the modify right

Paper 1 does not use the phrase “directed selection.” It does, however, establish the governance authority that makes directed selection possible: the modify right. Under Paper 1’s human-governed commitment, humans retain the right to modify any substrate content and any orchestration rule at any time during the substrate’s existence. This right is not qualified by the type of content being modified—it applies to the full substrate, including the DNA-layer content that Papers 2 and 3 operate on.

The architectural significance of this baseline is precise. When Paper 2 introduces directed selection as the named mechanism for governance-authorized DNA change, it does not introduce a new category of governance authority. Directed selection is an exercise of the modify right, scoped to DNA-layer content, performed under governance authorization, with a version record produced as output. Paper 1 already authorizes it; Paper 2 names and formalizes the mechanism that exercises that authorization in the evolution domain.

This means that directed selection—at any scope and with any source—never requires governance authority beyond what Paper 1 already establishes. An implementation of directed selection with inter-Self source content (Paper 3) does not need a new governance primitive; it exercises the same modify right Paper 1 commits to.

2.2 Paper 2: Explicit directed selection as the named DNA evolution mechanism

Paper 2 names directed selection as the primary governance-controlled evolution mechanism for DNA-layer content. Its defining properties are:

- **Governance authorization required.** A directed selection event is not self-executing. It requires explicit governance authorization—human authority exercising the modify right over a proposed DNA change.
- **Source is intra-Self action-layer content.** The improvement proposals that a directed selection event evaluates and acts on come from the proposing Self's own action-layer records, accumulated through the action-feedback mechanism. The Self's lived operational experience—what worked, what failed, what revealed gaps—is the content that human governance reviews and authorizes as DNA change.
- **Output is a versioned DNA change record.** Every directed selection event produces a DNA change with version record: the content of the change, its authorization, and its provenance metadata, all substrate content under the same governance rules that govern everything else.

At Paper 2 scope, these three properties exhaust the definition. The source is always intra-Self. The mechanism is what Paper 2 calls DNA evolution—governance-defined, goal-directed improvement of substrate content that distinguishes the architecture from undirected mutation.

2.3 Paper 3: Directed selection with inter-Self source content at the FAI absorption step

Paper 3 introduces Full Aspect Integration (FAI) as the canonical operation over the shared substrate. At FAI dissolution, content from the shared substrate propagates back to each participating Self's home substrate under governance-configured ingestion. For DNA-layer content, the propagation mechanism is stated explicitly in Paper 3: absorption operates through Paper 2's DNA evolution mechanism—directed selection under home governance.

This is directed selection with a different source. Instead of the proposing Self's own action-layer records, the source of the candidate DNA content is the contributed aspects of another Self—orchestration patterns, schemas, and rules that traveled through the shared substrate during the FAI event. The receiving Self's governance structure reviews and authorizes which of this content becomes a DNA change in the home substrate; what is

declined is not absorbed; what is approved is absorbed with provenance carried over at governance-configured depth.

Every property that defines directed selection in Paper 2 is present at the Paper 3 absorption step:

- Governance authorization is required. No DNA content from FAI is absorbed without explicit authorization from the receiving Self's authority structure.
- Output is a versioned DNA change record. The absorbed content enters the home substrate as a governed DNA change, with provenance metadata specifying that its origin is an FAI event from a contributing Self.
- The mechanism is governance-authorized DNA modification. Exactly the mechanism Paper 2 defines and Paper 1's modify right authorizes.

The only structural difference from Paper 2's directed selection is the source: inter-Self rather than intra-Self.

3. The disambiguation

The precise statement of the disambiguation is this:

Directed selection throughout the trilogy is the governance-authorized mechanism for modifying DNA-layer substrate content under human authority, producing versioned DNA change records. Across the three paper contexts, the mechanism, authorization requirement, and output type are identical. What varies is the source of the content under consideration.

Three source variants correspond to the three paper contexts:

- **Paper 1 (source unspecified):** The modify right authorizes governance to change any substrate content, including DNA content. No source is specified because Paper 1 operates at cell scope before the DNA/action layer architecture is named. The authority exists; the specific mechanism for its exercise in the evolution domain is Paper 2's contribution.
- **Paper 2 (intra-Self source):** Directed selection operates on improvement proposals derived from the proposing Self's own action-layer records. The source is home—the Self's accumulated operational experience.
- **Paper 3 (inter-Self source via FAI absorption):** Directed selection operates on DNA content contributed by another Self's aspects through the FAI shared substrate. The source is external—another Self's accumulated substrate content. Authorization, mechanism, and output type are unchanged.

What this disambiguation rules out is the reading that Paper 3's absorption protocol introduces a new evolution mechanism. It does not. Absorption is directed selection. The governance-authorized selective filtering of contributed DNA content, the version-recording of absorbed changes, the requirement for home-authority authorization before any content enters the home DNA layer—these are not properties of a novel mechanism; they are the properties Paper 2 already specifies for directed selection, applied to an inter-Self source.

The T2 subseries pattern this note establishes is: Paper 3's evolution concepts extend Paper 2's mechanisms to accept inter-Self source content. The mechanism is conserved; the source expands. This note establishes that pattern for directed selection. Subsequent T2 evolution disambiguation notes should follow the same analytical structure: identify the mechanism as defined in Paper 2, identify the source as it operates in Paper 3, and show that only the source differs.

4. Prior-art closure

Against the “absorption is a novel mechanism” adversary reading. Anyone implementing governance-authorized selective absorption of DNA content from another Self's contributed aspects through an inter-Self coordination event is implementing directed selection with inter-Self source content. The mechanism is prior art in Paper 2; the source expansion is prior art in Paper 3; the combination has been disclosed.

Against the “directed selection requires new Paper 3 authority” reading. The governance authority for directed selection with inter-Self source content is the modify right Paper 1 establishes. No new authority primitive is introduced at Paper 2 or Paper 3. Paper 2 names the mechanism that exercises that authority in the evolution domain; Paper 3 extends the sources from which that mechanism receives candidate content.

Against the “same mechanism in two papers must be different mechanisms” reading. The trilogy's architecture is explicitly designed so that Paper 2 mechanisms operate at intra-Self scope and Paper 3 extends them to inter-Self scope without replacing them. Directed selection is one instance of this design. The mechanism is one; the scope expands.

The unified claim. Directed selection in the CKS trilogy is the governance-authorized DNA change mechanism. Paper 3's DNA absorption step is directed selection with inter-Self source. The same mechanism operates in three contexts with three source configurations. No implementation of the absorption protocol—however it varies in provenance-carry-over depth, asymmetric ingestion configuration, or governance-configured filtering—constitutes a novel evolution mechanism relative to the directed selection architecture Paper 2 defines and Paper 1's modify right authorizes.